# Victorine Younang, Ph.D.

**Applied Scientist / Machine Learning Researcher — Ph.D., Computer Science**

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## SUMMARY

Applied ML Scientist (Ph.D., Computer Science) and Big Data Engineer specializing in foundation-model fine-tuning, Graph Neural Networks, and scalable real-time inference. Expert across the end-to-end ML lifecycle — from distributed Hadoop/PySpark data pipelines to curriculum fine-tuning stacks for MoE LLMs on constrained GPUs. Proven production impact (a graph recommender driving +33% session time at Rec Room); combines rigorous modeling (Bayesian methods, GNNs) with deep systems engineering (PyTorch, DDP, vLLM).

## TECHNICAL SKILLS

- **ML & Deep Learning:** PyTorch, Hugging Face, PyTorch Geometric; Graph Neural Networks (GAT/GNN), foundation models, transformers, BioBERT/ClinicalBERT, representation learning, anomaly detection
- **LLMs & Training:** LLM fine-tuning (LoRA/QLoRA, PEFT, MoE), RAG, DPO/RLHF, custom LLM evaluation & benchmarking, 4-bit NF4 quantization, gradient checkpointing, Distributed Data Parallel (DDP), vLLM serving
- **Big Data & Systems:** PySpark, Apache Spark, Hadoop/HDFS, MapReduce, Kudu, Elasticsearch, Impala, SQL; Docker, Kubernetes, MLflow, MLOps / model lifecycle, HPC/Slurm
- **Modeling & Eval:** Bayesian inference, VAR/time-series forecasting, ST-GNNs, adversarial robustness, responsible AI, ROC-AUC/PR-AUC, ablation studies; Python, R

## EXPERIENCE

### Oakland University | Applied ML Researcher

*Michigan, USA · Sep 2020 – Apr 2026*

- **LLM Fine-Tuning & Distributed Training (TIA):** curriculum fine-tuning of six foundation LLMs (incl. DeepSeek-V2 MoE, Qwen2.5) on constrained 16 GB GPUs via 4-bit NF4 quantization, gradient checkpointing, label-masked PEFT, and Distributed Data Parallel (DDP).
- **Graph Representation Learning (CryptoGraph):** hybrid Graph Attention Network + MLP in PyTorch Geometric over AST graphs; variant-aware stratified sampling + GraphNorm → F1 0.94 with robust zero-shot generalization.
- **Dynamic Probabilistic Modeling:** complex-valued Bayesian Attack Graph for real-time risk scoring; Vector AutoRegression (VAR) forecasting driving a dynamic allocation policy (+20% system lifetime, 95% request fulfillment).
- **Custom ML Evaluation:** neuro-symbolic metrics (syntactic validity, semantic match, value consistency) beyond BLEU/CodeBLEU.

### Health Alliance Plan (HAP) | Applied Scientist (NLP & Multimodal AI)

*Detroit, MI · Summer 2025*

- **Multimodal Representation Learning:** multimodal deep-learning system fusing BioBERT/ClinicalBERT text embeddings with structured tabular features to predict high-risk 30-day hospital readmissions.
- **Advanced NLP & Inference:** transformer NER pipeline extracting medical entities, social determinants of health (SDoH), and complex temporal negations from raw Epic EHR prose.
- **Deep-Learning Benchmarking:** PyTorch / scikit-learn frameworks benchmarking multimodal DNNs against tuned tree-based baselines (latency / Recall tradeoffs).
- **Scalable Deployment:** productionized the NLP + classification architecture as a modular, Dockerized microservice for healthcare management systems.

### Rec Room | Applied Scientist (Recommendation Systems)

*Seattle, WA · Summer 2023*

- **Graph-Based Recommendation Engine:** built and productionized a large-scale recommender (Node2Vec / random-walk embeddings) over a bipartite graph of 1,000+ virtual environments.
- **Real-Time Inference & Latency:** refactored the Python similarity-inference pipeline, –10% serving latency under peak concurrency; cold-start logic from structural graph properties.
- **Online Impact:** shipped to production → statistically significant +33% average user session time on first deployment.

### EDF (Électricité de France) | Applied ML Scientist (Graph ML & Time-Series)

*Paris, France · Summer 2021*

- **Spatial-Temporal GNNs:** designed and trained deep ST-GNN architectures in PyTorch capturing non-linear temporal + physical spatial dependencies across the low-voltage distribution grid.
- **Probabilistic Loss Engineering:** quantile / Pinball loss and Gaussian processes for strict 24-hour confidence intervals on renewable generation.
- **Physics-Informed DL:** custom loss regularizers penalizing physically impossible predictions (capacity limits, electrical topologies, Kirchhoff's laws).
- **Scalable Deployment:** Dockerized PyTorch inference + PySpark spatio-temporal preprocessing microservices for Smart Grid operations.

**EDF (Électricité de France) | Big Data & Machine Learning Engineer**
*Paris, France · 2019–2020*

- **Distributed Data Pipelines:** architected scalable ETL across Hadoop / MapReduce clusters processing millions of customer profiles for a consent-management platform.
- **Data Quality for ML:** engineered GDPR data-sanitization and algorithmic validation workflows, eliminating contamination and reducing downstream ML/analytics errors by 30%.

**Orange Telecom | Applied ML Scientist (Data Architecture & Forecasting)**
*Paris, France & Brussels, Belgium · 2017–2019*

- **Large-Scale ML Infrastructure:** architected the data topology for the HUB DATA France initiative, mapping application dependencies to optimize training-data flow into distributed Hadoop clusters.
- **Algorithmic Time-Series Forecasting:** multivariate predictive models in R forecasting Hadoop connection volumes and query loads → dynamic allocation of cluster compute during peak traffic.
- **Real-Time Telemetry:** high-throughput Python pipelines processing unstructured XML network logs into low-latency feature stores (Elasticsearch, Apache Kudu).
- **Distributed Compute Optimization:** authored an R-to-Impala connection package optimizing distributed storage/retrieval and standardizing the applied-research team's pipeline.

## SELECTED PROJECTS

- **RepoSage — Agentic, execution-verified RAG** (LangGraph, vLLM, Qdrant, DPO): dual-source code-&-docs RAG that generates, executes in a sandbox, and self-corrects against a library's own tests; DPO alignment from execution-feedback preferences.
- **CipherGuard — reference-free backdoor detection** (GAT + AST, Isabelle/HOL-verified corpus): catches test-vector-compliant backdoors with zero false positives on 44 ciphers (NDSS 2027).
- **NLP at Scale — BERT stance/sentiment classifier:** scraped 19,000+ posts; fine-tuned BERT to 96% accuracy with temporal topic modeling (Springer LNCS).

## EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (Sep 2020 – Apr 2026; GPA 3.96). Dissertation: *From Perceived Trust to Provable Security: Design, Optimization & Verification of Resource-Constrained IoT Ecosystems.*
- **M.Sc. Computer Engineering & B.Sc. Electronic & Computer Engineering** — ESIGELEC, France (2016–2019; GPA 3.87). *Big Data for Digital Transformation.*

## SELECTED PUBLICATIONS

- V. Younang et al. "Security Risk Assessment with Bayesian Attack Graphs and Complex Probabilities." *IEEE TDSC*, 2025.
- V. Younang et al. "Extending IoT Devices' Network Lifetime." *IEEE TSC*, 2025.
- V. Younang et al. "Comparative Analysis of Users' Concerns and Government Policies on Autonomous Vehicles." *Springer ICIOT / LNCS*, 2022.
- V. Younang et al. "CipherGuard: Reference-Free Detection of Backdoors in Cipher Implementations." *Under review, NDSS 2027.*
- V. Younang et al. "Tiered Isomorphic Alignment: Curriculum Fine-Tuning across Six LLMs." *Under review, AAAI 2027.*
- V. Younang et al. "CryptoGraph: Learning Security Properties of Block Ciphers with Graph ML." *In prep (resubmission), IEEE TIFS.*

## CERTIFICATIONS

- **AWS Certified AI Practitioner (AIF-C01)** — expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — expected Dec 2026; transformers, prompt engineering, fine-tuning, NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated in RepoSage.